

Yujie Zang

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EDUCATION

National University of Singapore (NUS)

Singapore

■ M.Sc. in Robotics

July.2026 – Present

- Selected Coursework: Robot Kinematics, Robot Dynamics and Control, AI Planning and Decision Making, Materials, Sensors, Actuators & Fabrication in Robotics

Shanghai Jiao Tong University (SJTU)

Shanghai, China

■ B.Eng. in Mechanical Industrial Design (Outstanding Graduate)

Aug.2022 – Jun.2026

• Grades: 90.10 / 100 | GPA: 3.91 / 4.30 | Ranking: 4 / 36

- Selected Coursework: Product Design, Materials and Structures, Advanced Mathematics, Computer Programming, Data Science, Machine Learning

RESEARCH INTEREST

My long-term research goal is to build robotic systems that leverage **human priors** to advance toward **superhuman physical intelligence**. In particular, I focus on **real-world robotic manipulation**, using **force and tactile sensing** to understand physical interactions, **world models** to predict their evolution, and **real-robot post-training** to acquire precise and fast skills.

PUBLICATIONS

[1] TacForeSight: Force-Guided Tactile World Model for Contact-Rich Manipulation.

Yujie Zang, Yuhang Zheng, Xian Nie, Yupeng Zheng, Shuai Tian, Songen Gu, Chen Gao, Zining Wang, Shuicheng Yan*, and Wenchao Ding*.

IEEE Robotics and Automation Letters (RA-L), 2026. (Under Second-Round Review)

[2] Multimodal Palpation Sensing for Precise Prediction of Breast Lump Hardness and Size.

Yujie Zang, Shuhong Xiao, Peng Yang, Ying Yang, Zixuan Lu, Ruyue Sun, Chongpeng Wei, Shuyue Zheng, Chen Wang, Xinyang Tan*, and Hongwei Zhang*.

IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2026. (Accepted)

[3] Re-DirectExo: A Preliminary Study of Illusion-Driven Hand Rehabilitation Exoskeleton.

Yujie Zang, Guiqin Lu, Zhaoyi Chen, Zhen Han, Zihang Ma, Xinyang Tan*, and Shi Qiu*.

International Conference on Human-Engaged Computing (ICHEC), ACM EI Short Paper, 2025. (Accepted)

[4] Generating Activity-Enriched Abnormal ECGs via Adversarial Network.

Liuqing Chen, Shuhong Xiao, Yujie Zang, Jiner Wang, Shanhui Hu, Lin Zhang, Qing Li, and Danyang Zhang.

IEEE Journal of Biomedical and Health Informatics (JBHI), 2025. (Accepted)

[5] OmniVTA: Visuo-Tactile World Modeling for Contact-Rich Robot Manipulation.

Yuhang Zheng, Songen Gu, Weize Li, Yupeng Zheng, Yujie Zang, Shuai Tian, Xiang Li, Ruihai Wu, Ce Hao, Chen Gao, Si Liu, Haoran Li, Yilun Chen, Shuicheng Yan*, and Wenchao Ding*.

IEEE Transactions on Robotics (T-RO), 2026. (Under Second-Round Review)

[6] VT-WAM: Visual-Tactile World Action Model for Contact-Rich Manipulation.

Shuai Tian, Yupeng Zheng, Yuhang Zheng, Songen Gu, Yujie Zang, Yuxing Qin, Weize Li, Haoran Li, Wenchao Ding*, and Dongbin Zhao*.

IEEE Robotics and Automation Letters (RA-L), 2026. (Under Review)

INTERNSHIP

Anyverse Dynamics

Beijing, China

■ Robotics Algorithm Intern | Advisor: Prof. Zhongpu Xia

Jun.2026 – Aug.2026

- **Topics:** Dual-Arm Manipulation Pipeline, Human-in-the-Loop Robot Learning, Real-Robot Reinforcement Learning

TARS Robotics

Shanghai, China

■ Robotics Research Intern | Advisors: Prof. Wenchao Ding and Prof. Zining Wang

Jan.2026 – Jun.2026

- **Topics:** Tactile Manipulation, Robot Learning, Tactile-UMI Hardware Design

COMPETITIONS & SKILLS

■ **Champion-SJTU**, RoboMaster University Championship

Sep.2024

■ **Third Prize**, China–U.S. Young Maker Competition (Shanghai Division)

July.2024

■ **Honorable Mention**, Mathematical Contest in Modeling (MCM)

May.2024

- **Robotics & Hardware:** SolidWorks, Mechanical Design, Machining & Assembly, Embedded Electronics, Sensor Integration, ROS/ROS2, Robot Kinematics, Real-robot Deployment

- **Robot Learning & Programming:** Reinforcement Learning, Imitation Learning, Diffusion Policy, Tactile Manipulation, Human-in-the-Loop Learning